FIGURE FORMATION

Assembling Pieces, Embedded and Not-Embedded Figures

OAVS TGT | Reasoning (Non-Verbal) | Topic 10: first non-verbal topic

Syllabus line (Reasoning, 30 Marks): “The Non-Verbal Reasoning topics like **Figure Formation**, Dice, Triangle, Rule Detection, Images, Completion of Pattern, Mirror Images, Figure Matrix and Paper Folding, etc.”

Built from: OAVS TGT English 2019, English 2022, Mathematics 2024, Science 2024 and Social Studies 2023 (17 May 2023, Shift 2). The 3 figure-formation questions found in these papers are solved here with their original figures. The practice figures were generated and **verified by computer search** so each question has exactly one correct option.

What is inside

Ch.	Title	What you get
1	PYQ Analysis	Counts, patterns, and a map of all 32 non-verbal PYQs to the nine topics
2	Foundation	Question types, rules and strategy for assembling and embedded figures
3	Solved PYQs	English 2019 Q39, SST 2023 Q18 and Q22 with figures
4	Shortcuts, Traps and Scope	Speed methods and common mistakes
5	Practice Set	12 original figure questions with a verified answer key
6	Quick Revision	One-page cheat sheet

PRIORITY: Only 3 of the 32 non-verbal PYQs are figure formation. The heaviest non-verbal topics are Mirror Images (8) and Rule Detection / figure series (7); the map in Chapter 1 shows the plan for the other topics.

CHAPTER 1

PYQ Analysis

Figure formation: 3 questions in the five papers; the non-verbal map

Paper	Q. numbers	Count	What was asked
English 2019	39	1	Which figure can be formed from the pieces in figure X
English 2022	none	0	—
Mathematics 2024	none	0	—
Science 2024	none	0	—
Social Studies 2023	18, 22	2	Embedded figure; figure NOT embedded (rotation not allowed)
Total		**3**	**Found in 2 of the 5 papers**

Map of all 32 non-verbal PYQs

Syllabus topic	PYQs	Count
Figure Formation (this PDF)	Eng 2019 Q39; SST 2023 Q18, Q22	3
Dice and cubes	Eng 2019 Q55; Eng 2022 Q40, Q53; Science 2024 Q96; SST 2023 Q13	5
Triangle (counting figures)	Eng 2022 Q32	1
Rule Detection (figure series / patterns)	Eng 2019 Q36; Maths 2024 Q75, Q96; Science 2024 Q79, Q98; SST 2023 Q7, Q26	7
Mirror Images	Eng 2019 Q34, Q46; Eng 2022 Q35; Maths 2024 Q77, Q79, Q93; Science 2024 Q97; SST 2023 Q4	8
Completion of Pattern	Eng 2019 Q56	1
Figure Matrix	Eng 2019 Q53; Eng 2022 Q47	2
Paper Folding (including transparent-sheet folds)	Eng 2019 Q47, Q48, Q49; Eng 2022 Q31	4
Venn diagram with numbers (not in the listed topics)	SST 2023 Q12	1
Total		**32**

PYQ INSIGHT: **Figure formation is a small part of the paper** (3 of 32 non-verbal questions). Heavier topics deserve more time: Mirror Images (8), Rule Detection (7), Dice and cubes (5), Paper Folding (4).

PYQ INSIGHT: **Embedded-figure questions say ‘Rotation is NOT allowed’.** The target must appear in the option exactly as drawn: same size, same orientation, no flipping.

CHAPTER 2

Foundation

Question types, rules and strategy

Type	Typical wording	You do
Assemble pieces	Find out which of the figures can be formed from the pieces given in figure X	Find the option that is exactly the pieces joined
Embedded figure	Select the option in which the given figure is embedded (rotation is NOT allowed)	Find the option that contains the figure unchanged
Not embedded	Select the option that is NOT embedded in the given figure	Find the option that cannot be found inside the main figure
Counting figures	How many triangles / squares are there?	Triangle topic (separate PDF)
Completing a figure	Which piece completes the pattern?	Completion of Pattern (separate PDF)

2.1 Assembling pieces: the rules

Rule	Meaning
1. Use every piece exactly once	No piece can be left out or used twice.
2. Shape and size cannot change	Edges, angles and lengths stay the same.
3. Rotation is allowed; flipping usually is not	Turn a piece freely; do not turn it over unless told.
4. No gaps and no overlaps	The formed figure is exactly the union of the pieces.
5. Matching edges	Two pieces join only along edges of equal length and a matching angle.
6. Outline = outer edges	The pieces' outer edges form the figure's border; inside lines are the joins.

Strategy

- **Count** pieces and note unusual edges (a slanted side, a notch, a right angle).
- **Find where the unusual edge fits**: a notch needs a piece with the same slanted edge.
- **Eliminate options** that contain extra lines or crossing diagonals that no piece can produce.
- **Check area**: the total area of the pieces must equal the area of the formed figure.

Worked example



Two pieces (an L of 3 squares and a domino) can be joined, edge to edge, into the 5-square figure on the right. The colours show where each piece sits. Every square of the formed figure belongs to exactly one piece.

2.2 Embedded figures: the rules

Rule	Meaning
1. Same orientation	No rotation and no flipping.
2. Same size	The figure's sides must be the same length as in the option.
3. Complete	Every line of the figure must be present in the option; extra lines may exist around it.
4. Lines are continuous	A line that is unbroken in the figure cannot be split in the option.

Strategy

- **Pick the most distinctive feature** of the figure (a circle, a curve, a sharp angle, a small box) and look for it in every option first.

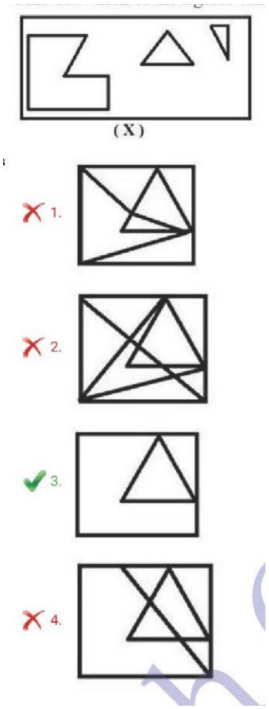
- **Check the surroundings** of that feature: what is attached to it and in which direction.
- **Reject an option at the first mismatch** (a box with extra bars, a changed end, a wrong proportion).
- **For 'NOT embedded'**, test each option's most distinctive feature against the main figure; the option whose feature is missing is the answer.

CHAPTER 3

Solved PYQs

The three figure-formation questions with their original figures

3.1 English 2019 Q39: which figure can be formed



1. The pieces in figure X are: **(a)** a large piece like a rectangle with a slanted notch cut from the top, **(b)** an equilateral triangle, **(c)** a small right-angled triangle (one vertical side).
2. A figure formed from them must be exactly these three pieces joined, with no extra lines, gaps or overlaps.
3. Options 1, 2 and 4 contain extra crossing diagonals and overlapping triangles that three pieces cannot produce.
4. Option 3 is a plain rectangle. The large piece is its left and bottom part. The notch's slanted edge matches the **left side of the triangle**, and the notch's horizontal edge matches the triangle's **base**. The small right-angled triangle fills the gap between the triangle's right side and the rectangle's right edge. All pieces used once, nothing left over.
Answer: Option 3

SHORTCUT: Method: locate the odd edge (the slanted notch), find which piece has the same slanted edge (the triangle), then see what space remains (the small triangle). A rectangle results, so the matching option is the plain rectangle.

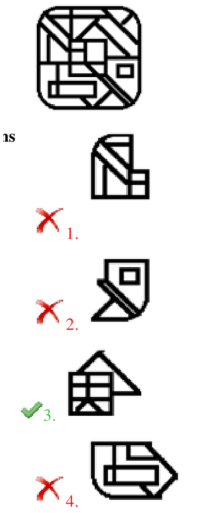
3.2 SST 2023 Q18: embedded figure



1. The target (the figure at the top) has: a **small circle on a stem**, a **square box on the left**, a **square box on the right**, and a long **tail that ends in a point** below the stem.
2. Start with the most distinctive feature, the circle on the stem, and then check the neighbours.
3. Option 4 contains the circle on the stem, both boxes in the same positions and proportions, and the pointed tail, hidden among extra lines.
4. In the other options at least one feature is changed: in option 1 the boxes are flat wide rectangles and the tail ends in a flat bar; in option 3 the right box has vertical bars inside it; in option 2 the boxes are cut by extra lines.
Answer: Option 4

NOTE: The scanned source figure is small, so the elimination notes for options 1 to 3 are read from the scan. The answer (Option 4) is the official key.

3.3 SST 2023 Q22: figure NOT embedded



1. The main figure (top) is a **square with rounded corners** on the outside, filled with diagonal bands, boxes and bars.
2. Test each option's distinctive feature: option 1 has a **rounded top-left corner**, option 2 a **rounded bottom-right corner**, option 4 a **rounded bottom-left corner**. Each of these matches a rounded corner of the main figure.
3. Option 3 has **no rounded corner**; it has a pointed triangular roof above a grid of small rectangles. The main figure has no such roof or grid, so option 3 is not embedded.
Answer: Option 3 (the figure that is NOT embedded)

SHORTCUT: For 'NOT embedded' questions, three options **are** in the figure. Find a feature that identifies where each of the three sits (here, which corner), and the fourth has none.

CHAPTER 4

Shortcuts, Traps and Scope

Speed methods and common mistakes

- **Assembling:** match the slanted or notched edge first, then check what remains.
- **Eliminate extra lines:** options with lines no piece can supply are wrong.
- **Embedded:** check the most unusual feature (curve, circle, sharp angle) in all four options before anything else.
- **NOT embedded:** three options have a feature you can place in the main figure; the fourth does not.
- **No rotation:** if a figure is turned, it is a different figure.

Trap	How to avoid it
Rotating the target to make it fit	'Rotation is NOT allowed': the target must match as drawn.
Accepting a figure with a broken line	The target's continuous lines must not be split in the option.
Ignoring size	A similar shape of different proportions is not the target.
Using pieces twice or leaving one out	Every piece appears once in the formed figure.
Choosing the busiest option for 'embedded'	More lines do not mean the target is inside; check the features.

CHAPTER 5

Practice Set

12 original figure questions; answers on the next page

Rotation of pieces is allowed in Part A; pieces may not be flipped. In Parts B and C figures may not be rotated or flipped (only moved).

Part A: Which figure can be formed from the pieces?

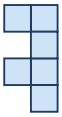
Q1. Pieces:




(A)	(B)	(C)	(D)
			

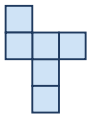
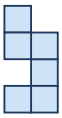
Q2. Pieces:



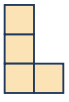

(A)	(B)	(C)	(D)
			

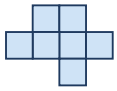
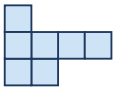
Q3. Pieces:




(A)	(B)	(C)	(D)
			

Q4. Pieces:

(A)	(B)	(C)	(D)
			

Part B: In which option is the given figure embedded?

Q5. Given figure:



(A)	(B)	(C)	(D)
			

Q6. Given figure:



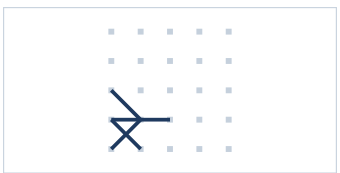
(A)	(B)	(C)	(D)
			

Q7. Given figure:



(A)	(B)	(C)	(D)
			

Q8. Given figure:



(A)	(B)	(C)	(D)
			

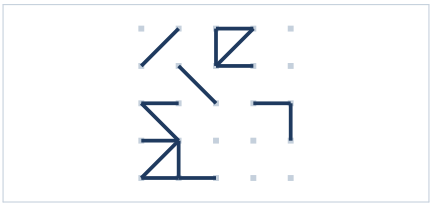
Q9. Given figure:



(A)	(B)	(C)	(D)

Part C: Which option is NOT embedded in the main figure?

Q10. Main figure:



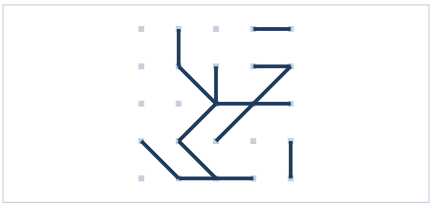
(A)	(B)	(C)	(D)

Q11. Main figure:



(A)	(B)	(C)	(D)

Q12. Main figure:



(A)	(B)	(C)	(D)

Answer key (verified by computer search)

Q	Answer	Solution
1	(B)	
2	(B)	
3	(C)	
4	(B)	
5	(D)	
6	(A)	
7	(A)	
8	(D)	
9	(B)	
10	(C)	Option (C) has a line pattern that cannot be placed anywhere in the main figure; the other three can.
11	(B)	Option (B) has a line pattern that cannot be placed anywhere in the main figure; the other three can.
12	(A)	Option (A) has a line pattern that cannot be placed anywhere in the main figure; the other three can.

KEY: **Key:** in Part A the colours show how the pieces fit. In Part B the embedded figure is shown in red inside the correct option. For Part C, each of the other three options appears (as drawn) somewhere inside the main figure.

CHAPTER 6

Quick Revision

One page for the last day

Question	Method	PYQ check
Which figure can be formed from the pieces	Match the odd edge; no gaps, no overlaps; reject extra lines	Eng 2019 Q39: plain rectangle (option 3)
Embedded figure	Most distinctive feature first; no rotation; all lines present	SST 2023 Q18: option 4
NOT embedded	Place each option's feature in the main figure; the one that fails	SST 2023 Q22: option 3 (no rounded corner)

ROUTINE: **Exam-day routine:** name the unusual feature, find it in the options, reject at the first mismatch. 40 seconds per question. Next in the syllabus order: Dice, Triangle, Rule Detection, Images, Completion of Pattern, Mirror Images, Figure Matrix, Paper Folding.